AP BIOLOGY TOPIC 7.4

Population Genetics: Drift, Gene Flow & Mutation

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.H | **EVO-1.I** | **GENETIC DRIFT & GENE FLOW**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.4 (POPULATION GENETICS: DRIFT, GENE FLOW & MUTATION)

CED Core Vocabulary

Gene Pool — The total collection of all alleles for all genes present in a reproducing population at a given time.

Evolution Includes Random Processes — Evolution is driven by both non-random natural selection and random (non-selective) processes — mutation, genetic drift, and gene flow — which alter allele...

Genetic Drift — Unpredictable, random fluctuations in allele frequencies from one generation to the next due to chance sampling error during reproduction. Strongest in SMALL...

2. Founder Effect — Occurs when a small group of individuals separates from a larger population to colonize a new, isolated habitat (e.g., an island). The founding group carries...

Allele Frequency = Measure of Evolution — Microevolution is defined as a change in allele frequencies in a population's gene pool over generations .

Mutation — Random changes in DNA nucleotide sequences; the ultimate source of all new alleles and genetic variation.

1. Bottleneck Effect — A sudden, severe reduction in population size (typically caused by a random environmental catastrophe such as wildfire, volcanic eruption, flood, or...

Bottleneck — Northern elephant seals hunted down to ~20 individuals in the 1890s → today's 150,000+ seals are nearly genetically identical at many loci.

§2. GENE POOLS & NON-SELECTIVE EVOLUTIONARY FORCES

Microevolution

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Gene Pools & Non-Selective Evolutionary Forces ➔ **Gene Pool** ➔ **Allele Frequency = Measure of Evolution** ➔ **Evolution Includes Random Processes** ➔ **Mutation**

- **Gene Pool:** The total collection of all alleles for all genes present in a reproducing population at a given time.
- **Allele Frequency = Measure of Evolution:** Microevolution is defined as a **change in allele frequencies in a population's gene pool over generations**.
- **Evolution Includes Random Processes:** Evolution is driven by both **non-random natural selection** and **random (non-selective) processes** — *mutation, genetic drift, and gene flow* — which alter allele frequencies regardless of whether an allele helps or harms fitness.
- **Mutation:** Random changes in DNA nucleotide sequences; the **ultimate source of all new alleles** and genetic variation.

§3. GENETIC DRIFT: EVOLUTION BY RANDOM CHANCE

Small Populations

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Genetic Drift ➔ **Genetic Drift** ➔ **1. Bottleneck Effect** ➔ **2. Founder Effect**

- **Genetic Drift:** Unpredictable, **random fluctuations in allele frequencies** from one generation to the next due to chance sampling error during reproduction.
  - **Strongest in SMALL populations** (sampling error has huge impact).
  - Reduces genetic variation over time → can cause random **fixation (** **100%** **)** or **loss (** **0%** **)** of alleles, even deleterious ones!
- **1. Bottleneck Effect:** A sudden, severe reduction in population size (typically caused by a random environmental catastrophe such as wildfire, volcanic eruption, flood, or overhunting).
  - The surviving gene pool is a small, random sample unrepresentative of the original population, resulting in severe **loss of genetic diversity**.
- **2. Founder Effect:** Occurs when a small group of individuals separates from a larger population to colonize a new, isolated habitat (e.g., an island).
  - The founding group carries only a fraction of the parent gene pool, so allele frequencies in the new population often differ dramatically from the source population.

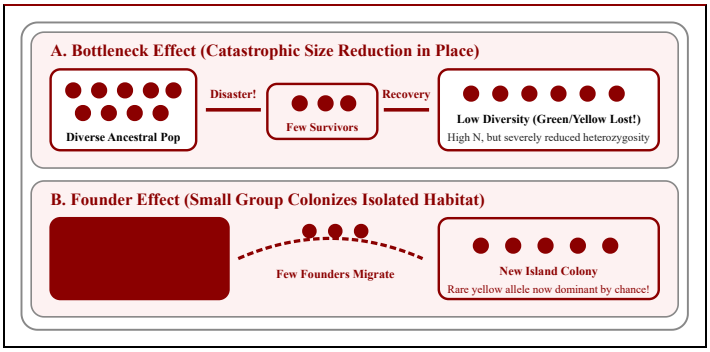
[COMMON AP EXAM FRQ TRAP] **Drift is NOT based on fitness — it is purely chance!** Do not say organisms survived a bottleneck because they were "more adapted" if the cause was a random disaster.

[EXAMPLE & BIOLOGICAL APPLICATION] **Drift Examples:**

- **Bottleneck:** Northern elephant seals hunted down to ~20 individuals in the 1890s → today's 150,000+ seals are nearly genetically identical at many loci.
- **Founder Effect:** Old Order Amish in Pennsylvania (founded by ~200 immigrants) have an unusually high frequency of recessive **Ellis-van Creveld syndrome** (polydactyly).

§5. VISUALIZING BOTTLENECK EFFECT VS. FOUNDER EFFECT

Genetic Drift Models



§6. SUMMARY: EVOLUTIONARY MECHANISMS CHANGING ALLELE FREQUENCIES

Comparison Matrix

MECHANISM	RANDOM OR ADAPTIVE?	EFFECT ON GENETIC VARIATION
Mutation	**Random** (with respect to fitness)	**Increases variation** by creating brand-new alleles in the gene pool.
Genetic Drift *(Bottleneck / Founder)*	**Random** (chance sampling error; strongest in small **N**)	**Decreases variation** within a population (causes allele loss or fixation); increases divergence between populations.
Gene Flow *(Migration)*	**Random / Non-adaptive** (transfers alleles between pops)	**Increases variation** within recipient population; **reduces differences** (homogenizes) between populations.
Natural Selection	**Non-Random & Adaptive** (Only force that produces adaptations!)	Increases frequency of beneficial alleles; decreases harmful alleles (can increase, decrease, or maintain variation).

[AP EXAM FRQ SCORING STRATEGY] **Hardy-Weinberg Preview:** A population is in **Hardy-Weinberg Equilibrium (no evolution)** ONLY when none of these forces act: (1) No mutation, (2) No gene flow/migration, (3) Very large population size (no drift), (4) Random mating, and (5) No natural selection!

§4. GENE FLOW (MIGRATION BETWEEN POPULATIONS)

Homogenizing Force

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Gene Flow (Migration Between Populations) ➔ **Gene Flow** ➔ **Evolutionary Effects**

- **Gene Flow:** The movement of alleles into or out of a population via the **migration of fertile individuals or gametes** (e.g., wind-blown pollen, animal dispersal) followed by interbreeding.
- **Evolutionary Effects:**
  - **Increases genetic diversity** within the recipient population by introducing new alleles.
  - **Acts as a homogenizing force:** Makes two connected populations *more genetically similar* over time, counteracting local natural selection and genetic drift.
  - **When Gene Flow Is Reduced/Blocked:** Populations become genetically isolated → promotes genetic divergence and potential **speciation**.

[CORE AXIOM / REMEMBER] **Gene flow = mixing gene pools.**

[COMMON AP EXAM FRQ TRAP] **Migration alone ≠ gene flow!** Moving into a new territory only counts as gene flow if the migrants **successfully interbreed/reproduce** and contribute alleles to the recipient gene pool.

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.4 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.4: Population Genetics: Drift, Gene Flow & Mutation ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE$_{\bar{x}}$)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE$_{\bar{x}}$):** If the ±2 SE$_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.4 to prove why the claim holds.

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